Deep inelastic scattering on polarized lithium: the physics behind the planned measurements

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Polarized lithium beams would provide the EIC with spin-polarized *nuclei* that are light enough for ab initio computation yet complex enough to exhibit genuine nuclear structure. ^{6}Li is a spin-1 nucleus well described as a deuteron bound to an inert α core; ^{7}Li is its spin-3/2 partner, an α plus a triton. This note develops, starting from the DIS formalism, the four measurements around which the program is built: the tensor structure function b_1, the double-helicity-flip ("exotic glue") structure function Δ, the polarized EMC effect, and spectator-tagged DIS on the embedded deuteron. It begins with the case itself: why lithium is the right element, why the observables exist only with polarization, and which of them a fixed target could deliver and which require the collider.

1 · Why polarized lithium, and why at a collider

The EIC project scope provides polarized electrons, protons and ^{3}He; polarized deuterons are planned as a later addition, and no heavier polarized ion appears in the conceptual design or in the Yellow Report [29,25]. The EPIOS white paper [1] makes the case for polarized ^{6}Li and ^{7}Li — "a desirable addition" that it lists with ^{2}H and ^{3}He as critical technology — and the ANL source development [51] specifies the beam: an atomic-beam source with optical pumping at 671 nm feeding EBIS charge breeding, with source targets $P_z \geq 0.90$ (vector) and $P_{zz} \geq 0.80$ (tensor) at $\geq 10^{13}$ atoms/s. Four measurements motivate the beam, each addressing a verified gap in the literature: no b_1 prediction or measurement exists for any nucleus beyond the deuteron; no gluon-transversity projection exists for any EIC target; no study of α-tagged DIS on lithium exists; and no lithium-beam luminosity or polarization parameter set has been published. The purpose of the program's simulations is to convert these gaps into figures of merit.

MEASUREMENT	OBSERVABLE	ELECTRON BEAM	ION SPIN	AXIS
Polarized EMC effect (^{7}Li, ^{6}Li)	$A_{\parallel} \to g_1^A$	polarized, helicity flipped	vector P_z	longitudinal
Tensor structure function (^{6}Li)	$A_{zz} \to b_1$	unpolarized	tensor P_{zz}	longitudinal (or along **q**)
Exotic glue (^{6}Li; ^{7}Li eligible)	$\cos 2\varphi \to \Delta(x,Q^2)$	unpolarized	tensor P_{zz}	**transverse**
Tagged DIS (both isotopes)	tagged versions of all three	as above	as above	as above, correlated with spectator

1.1 · Why lithium as a source: an alkali atom with the right nuclei

Lithium is the lightest alkali metal and its atom is hydrogen-like: a single 2s valence electron whose D lines near 671 nm (670.96 and 670.98 nm in vacuum for ^{7}Li) fall within the range of commercial single-frequency diode lasers, and whose ground-state hyperfine splittings, 228.2 MHz for ^{6}Li and 803.5 MHz for ^{7}Li [35], are small enough for one laser with an electro-optic sideband to address both hyperfine levels. Circularly polarized light pumps the atoms into a single magnetic substate, the hyperfine coupling transfers the electron polarization to the nucleus, and radio-frequency transitions between Zeeman sublevels then set any desired combination of P_z and P_{zz} [51]. The EPIOS paper makes the general point: alkali atoms "are hydrogen-like with a single valence electron and laser cooling of these species has been extensively studied", whereas boron, the other light candidate with a large nuclear spin, has an electronic structure that "presents a challenge for optical pumping" [1]. The low ionization energy of lithium (5.39 eV [35]) eases ionization, and fully stripped Li^{3+} is within reach of the EBIS that is being commissioned to charge-breed the polarized $^3He^{++}$ beam, whose design goal is 2×10^{11} ions per pulse at ≥ 70% polarization, with the measurement of the polarization transmitted through the EBIS planned for 2026 [34,1].

The nuclei are the other half of the argument. Both isotopes are stable and available (7.6% ^{6}Li and 92.4% ^{7}Li in natural lithium), and they are the only stable isotopes of one element with J = 1 and J = 3/2: the stable spin-1 nuclei are exactly ^{2}H, ^{6}Li and ^{14}N, and the stable spin-3/2 nuclei below A = 30 are ^{7}Li, ^{9}Be, ^{11}B, ^{21}Ne and ^{23}Na [36]. One source, one laser system and one injector therefore serve a spin-1 and a spin-3/2 program. The EPIOS assessment of where an exotic-glue discovery is most likely — "spin-1 ^{6}Li or ^{14}N, or spin-3/2 ^{7}Li or ^{23}Na" [1] — reduces, once optical pumping is required, to lithium and sodium, and lithium is the lighter, cluster-structured and ab initio-computable one (§1.2).

Polarized lithium beams also have a forty-year history at low energy, so the source physics is established rather than speculative. The Heidelberg tandem delivered $^6Li^{3+}$ beams of up to 150 nA with vector polarization 0.57–0.65 in 1977 [30]; by 1987 its atomic-beam source reached P_z, $P_{zz} \geq 0.85$ and accelerated ^{7}Li with an alignment $P_{zz} = 0.69$ on target [31]; the optically pumped Florida State source delivered more than 150 nA of $^6Li^{3+}$ with $P_{yy} = -1.33 \pm 0.02$ and $P_y > 0.74$, with no polarization loss in the linac [32]; and polarized ^{6}Li, ^{7}Li and ^{23}Na beams served two decades of nuclear-reaction studies [33]. What is new for the EIC is scale and transport: $\geq 10^{13}$ atoms/s, two to three orders of magnitude beyond these sources, and the preservation of the polarization through charge breeding and acceleration to more than 100 GeV/u [51,1].

In the ring, ^{6}Li is the easy case. Its gyromagnetic anomaly, $G \approx -0.18$, is as small as the deuteron's (−0.14), so it crosses only 81 depolarizing resonances on the way to top energy, against 63 for the deuteron and 2457 for ^{3}He [28], and, like the deuteron, it is handled with a partial solenoid snake and tune jumps — full Siberian snakes would be impractically long for so small a G [28,1] — so the deuteron techniques [52] carry over. ^{7}Li ($G \approx +1.52$) crosses 573 resonances and needs partial snakes with jump quadrupoles [1,28]. The rigidity of the hadron ring fixes the top energies at 275 GeV × Z/A: 137.5 GeV/u for ^{6}Li and 117.9 GeV/u for ^{7}Li. Because ^{6}Li has A/Z = 2, it runs at the same per-nucleon energy and magnet settings as the deuteron, so e+d and e+^{6}Li can be compared at identical kinematics.

1.2 · Why lithium as a target: the spin sits on a light cluster

The two stable lithium isotopes are the lightest nuclei beyond A = 3 in which a polarized light cluster is bound to an inert core. In ^{6}Li the core is an α particle (J = 0, T = 0) and the cluster is deuteron-like, bound by 1.47 MeV against α+d breakup; in ^{7}Li the cluster is a triton, bound by 2.47 MeV [36]. Variational Monte Carlo calculations with realistic two- and three-nucleon forces make the picture quantitative: the α–d overlap of the ^{6}Li ground state integrates to $N_{\alpha d} = 0.86$, of which 0.846 is S-wave and 0.017 D-wave, and the α–t overlap of ^{7}Li integrates to $N_{\alpha t} = 1.00$ [17]; the no-core shell model with continuum gives an asymptotic D/S ratio of −0.027 for the d+α configuration [38]. The consequence for polarization is what makes the isotopes useful. In a fully polarized ^{6}Li the α contributes nothing and the two valence nucleons carry the spin: six-body VMC gives spin-up and spin-down valence numbers of 1.924 and 1.076, a polarization of 0.85 [17,19]; three-body α+n+p models, in which the α+d component accounts for 60–70% of the wave function, give a valence-nucleon polarization above 90% [18]; and the SLAC E155 analysis, which used ^{6}LiD as its deuteron target, adopted 87% for the effective deuteron polarization from the same body of calculations [43]. ^{6}Li is therefore a polarized deuteron embedded in the field of an α particle — a spin-1 target whose tensor observables can be set against those of the free deuteron to isolate the medium effect. The only other stable spin-1 ground state is ^{14}N, whose spin is shared among p-shell nucleons, which has no comparable cluster simplicity, and which lies beyond the reach of exact quantum Monte Carlo ($A \leq 12$) [40].

In ^{7}Li the spin is carried by the unpaired proton of the triton cluster, the neutrons being paired [39]. VMC gives spin-up and spin-down proton numbers of 1.934 and 1.066 and neutron numbers of 1.981 and 2.019 [17], i.e. $P_p = 0.87$ and $P_n = -0.04$ — quoted as 0.866 and −0.037 in the JLab proposal [45] and used in §4.3. The shell model already gives $P_p = 13/15$; the microscopic calculations confirm the proton value and show the neutrons almost unpolarized [19,14]. Polarized ^{7}Li is thus to the proton what polarized ^{3}He is to the neutron, with the property, decisive for the polarized EMC effect, that the proton sits in a nucleus of normal density. Cloët, Bentz

and Thomas state the selection criteria: the nucleus must have $|P_n| \ll |P_p|$, and it must be light, because g_1^A is carried by the valence nucleons and is suppressed relative to F_2^A by roughly 1/A; their candidates are ^{11}B, ^{15}N, ^{27}Al and ^7Li, "where the nuclear polarization is largely dominated by the valence proton" [14], and the gluon-spin study names ^7Li "the most promising case for experimental investigation" [16].

Two further properties matter. The cluster structure makes the nuclei computable: Green's-function Monte Carlo with the AV18+IL7 Hamiltonian reproduces the binding energies of both isotopes (−31.93 vs −31.99 MeV for ^6Li, −39.15 vs −39.25 MeV for ^7Li), their charge radii to within 3%, and, once two-body currents are included, their magnetic moments (0.835 vs 0.822 μ_N; 3.24 vs 3.256 μ_N) [40,39], so the nuclear corrections entering every extraction — effective polarizations, momentum distributions, spectral functions — carry controlled uncertainties rather than model spread. The quadrupole moment of ^6Li is the one exception: the measured Q = −0.0806 fm², against +0.286 fm² for the free deuteron [36], arises from a delicate cancellation between the deuteron's own quadrupole moment and the D-wave part of the α–d relative motion [19,37], and theory reproduces only its smallness. That smallness is itself an asset. ^6Li is nearly spherical, so tensor effects of geometric origin (the deformation term of the coherent channel, §4.2) are bounded and a tensor signal must be dynamical in origin; ^7Li, whose p-shell triton orbit gives Q = −4.00 fm² [36], is the deformed counterpart. Finally, the two isotopes share one source, one injector and one detector: the α+d versus α+t comparison changes only the cluster, the cleanest controlled variation the program has.

1.3 · Why polarization is indispensable

An unpolarized lithium beam would be one more light ion. Its inclusive cross section is carried by F_1 and F_2 alone — for an unpolarized target the spin-1 cross section has exactly the spin-½ form [2] — and it would add an A = 6, 7 point to the EMC and shadowing curves that ^4He and ^{12}C already map. Every item of the program is instead a spin observable, and the two flagship items need the tensor moment specifically. b_1 is *defined* as the difference between the m = 0 and m = ±1 quark distributions; averaging over m erases it, and it can be reached only with tensor polarization — which is why the only existing measurement is HERMES, with the only tensor-polarized target ever used in DIS [5]. Δ is generated by gluon transversity, an interference between gluon helicity states that requires a target spin of at least 1 [3], and its cos 2φ signature needs a transverse alignment axis to define the azimuth: the inclusive cross section of an unpolarized target has no azimuthal dependence at all. The polarized EMC effect is by construction a statement about g_1, and the gluon-spin EMC effect enters only through the Q^2 evolution of g_1 [16]. In tagged DIS the spectator momentum selects S- versus D-wave configurations of the polarized pair, removing depolarization at small p_T and producing tensor asymmetries of order unity near 300 MeV/c [41]: the spin is the signal. Since the EIC project polarizes only protons and ^3He, with deuterons to follow [29], polarized lithium would be the first polarized ion at a collider with a nuclear core — the first place where the spin structure *of a nucleus*, rather than of a nucleon, is studied at collider kinematics. Tensor polarization is the scarcer resource (source targets $P_{zz} \geq 0.80$ against $P_z \geq 0.90$ [51], with ring transport still to be established), and both b_1 and Δ depend on it.

1.4 · Fixed target or collider?

Polarized lithium has existed for decades as a fixed target, but only in the solid state. Dynamically polarized ^6LiD was the deuteron target of SLAC E155 and of COMPASS, ^7LiH has been polarized at Saclay (47% for ^7Li, 56% for the protons) [44], and dynamically polarized ^7LiD (with ^6LiH or NH_3 as the proton reference) is the material of the approved CLAS12 polarized-EMC experiment [43,42,45]. In these compounds the lithium polarizes along with the hydrogen isotope because all spin species share one spin temperature: COMPASS reached deuteron polarizations of +54%/−47%, and the equal-spin-temperature relation gives 51 ± 3% for ^6Li and 92 ± 4% for the ^7Li admixture [42]; E155 measured the ^6Li polarization at 97% of the deuteron's [43]; in ^7LiD at 5 T the CLAS12 proposal expects 65–80% for ^7Li [45]. Three limits are structural. (i) Dilution: only a fraction f ≈ 0.35–0.45 of the nucleons in ^6LiD are polarizable once the helium coolant is counted — better than 0.22 for ND_3 and 0.15 for NH_3, but far from the 1.0 of a stored beam [42,43]. (ii) Tensor polarization: in

thermal equilibrium the spin-1 tensor moment follows from the vector one, $P_{zz} = 2 - \sqrt{(4 - 3P_z^2)}$, so $P_z = 0.5$ gives $P_{zz} = 0.20$; radio-frequency "hole burning" has raised it to 29–37% in deuterated butanol at the cost of the vector polarization, and the JLab b_1 experiment is conditionally approved on demonstrating 30% [8,46]. An atomic-beam source prepares the tensor moment directly: HERMES ran its gas target at $P_{zz} = +0.89$ and -1.65 in the pure-tensor states, "a combination which is not possible for solid-state polarized targets" [5], and the lithium source targets $P_{zz} \geq 0.80$ [51]. (iii) Spectator loss: a spectator α of 100–300 MeV/c carries only 1.3–12 MeV in the target rest frame and stops within 10–250 μm of lithium deuteride (NIST ASTAR ranges [49]) — inside target cells 2–5 cm long at JLab and 30–60 cm long at COMPASS — and a coherent ^{6}Li recoil at $|t| = 0.05$ GeV^2 has only 4.5 MeV. The JLab recoil detectors that do tag slow spectators, BoNuS (protons from 70 MeV/c) and ALERT (α from 230 MeV/c), work with unpolarized gas at a few atmospheres [47,48]; polarized-lithium targets outside dynamically polarized solids are essentially unknown — a 1975 polarized-^{6}Li target for a low-energy reaction study [50] is the one instance on record — and none has been used in lepton scattering.

The collider removes each limit at once. The beam is pure lithium with its polarization prepared atom by atom and measured in the ring; the spectator α or triton carries the beam momentum per nucleon and emerges at a few milliradians into the Roman Pots, off-momentum detectors, B0 and zero-degree calorimeter, with ~10% transverse-momentum resolution at 100 MeV/c [22]; the intact coherent recoil can be tagged down to $p_T \approx 0$ with the IR-8 secondary focus [27]; and the centre-of-mass energy — $\sqrt{s_{NN}}$ up to 99 GeV for e(18 GeV) × ^{6}Li(137.5 GeV/u) — opens x down to ~10^{-4} at $Q^2 \geq 1$ GeV^2, where JLab at 11 GeV stops at $x \approx 0.06$ and HERMES stopped at 0.01 [45,5]. That is the region where gluons dominate, where HERMES saw its large b_1, and where coherence holds. What stays with the fixed target is luminosity: 10^{35} cm^{-2} s^{-1} with dynamically polarized targets (CLAS12 at 10–15 nA, Hall C at 115 nA, with the material annealed every 2–3 days) [45,8] against an electron–nucleon luminosity of 10^{33}–10^{34} cm^{-2} s^{-1} at the EIC [29,25]. The high-x polarized EMC effect and the high-x b_1 of the deuteron are therefore rightly JLab measurements, and the EIC program concentrates on what the table summarizes: the low-x and high-Q^2 regions, the tensor observables at full polarization, and everything that needs a detected spectator or recoil.

MEASUREMENT	AT A FIXED TARGET	WHAT ONLY THE COLLIDER ADDS
Polarized EMC effect, ^{7}Li (g_1^A)	Approved: JLab E12-14-001 (CLAS12, polarized ^{7}LiD, $0.06 < x < 0.8$, $1 < Q^2 < 15$ GeV^2, $P_z \approx 0.8$ assumed, 55 days) [45]	$x < 0.06$; the Q^2 evolution that carries the gluon-spin EMC effect [16]; an undiluted beam; α-tagging to identify the struck triton and control its virtuality (IP6 Roman Pots)
Tensor b_1, ^{6}Li	Deuteron only: E12-13-011 (ND_3, $0.16 < x < 0.49$, $P_{zz} \geq 0.3$ required) [8]; ^{6}LiD would give $P_{zz} \approx 0.2$ with the deuteron term to be subtracted — no such measurement has been proposed	x down to 10^{-3}, where HERMES found $b_1 \approx 0.1$; pure-tensor fills at $P_{zz} \approx 0.6$–0.8; spin axis along **q** [26]
Exotic glue Δ, ^{6}Li (^{7}Li)	Letter of intent only: LOI12-16-006 (^{14}N in NH_3, $x < 0.3$, 2–12% alignment, in-plane spectrometers — cos 2φ "difficult with a fixed target") [13]	Transverse tensor polarization ≥ 0.6 with full 2π azimuthal acceptance; the gluon-dominated region $x \lesssim 0.1$ and a Q^2 lever arm
Spectator-tagged DIS (α, t)	Not feasible: spectators stop within 0.25 mm of a solid target; no polarized lithium gas target exists	Far-forward detection of α and t at beam velocity; pole extrapolation to the free cluster [20,22]
Coherent intact-^{6}Li channel	Not feasible: needs $x \lesssim 0.01$ and a 2–9 MeV recoil	Roman-Pot p_T tail or the IR-8 secondary focus [27]

1.5 · What each observable requires of the beam, and what lithium alone provides

The requirements below are the program's own, from the fast simulation and the polligen pseudo-experiments (§5), stated for the placeholder in-ring polarizations $P_z = 0.7$ and $P_{zz} = 0.6$, $P_e = 0.7$, and 10 fb^{-1}/u. Every statistical error scales as $1/(P\sqrt{L})$, so a lower polarization is paid for in luminosity squared: halving P_{zz} costs a factor four in running time. Two machine facts shape the list. The stable spin direction in the arcs is vertical, so transverse polarization at the interaction point is native while longitudinal polarization needs spin rotators, whose lithium settings are not yet defined [29,1]; and tensor fills need m = 0 bunches prepared by the source

radio-frequency transitions, whose survival through acceleration and bunch-by-bunch patterning are open questions [51]. The observables also demand different spin configurations and therefore share the luminosity; the reach quoted assumes each receives the full 10 fb^{-1}/u in its own configuration. (Semi-inclusive channels add the three leading-twist tensor TMDs of a spin-1 target [1]; they need the same tensor polarization and are not projected here.)

OBSERVABLE	BEAM POLARIZATION REQUIRED	ELECTRONS	REACH AT 10 FB^{-1}/U	BINDING REQUIREMENT
g_1^A, polarized EMC (^7Li; ^6Li for the isoscalar pair)	Vector $P_z \geq 0.7$, **longitudinal** (spin rotators). A spin-3/2 spin-temperature fill at $P_z = 0.8$ also carries $P_{zz} \approx 0.55$ and $P_{zzz} \approx 0.11$; the octupole term enters the asymmetry numerator and is corrected through the fill model [45]	Polarized, $P_e \approx 0.7$, helicity flipped bunch by bunch	$\delta\Delta R \approx 2.6$–4% per x-bin at x = 0.3–0.5 (three energies combined); CBT-vs-TMT ≈ 1.5σ per bin, 5σ at x ≈ 0.5–0.7 with 100 fb^{-1}/u	Polarimetry scale $\delta P_z/P_z \leq 3\%$; P_p from VMC [17]; Q^2 lever arm at fixed x (gluon-spin EMC)
b_1 via A_{zz} (^6Li)	**Tensor** P_{zz}, longitudinal or along **q** [26]; fills of different P_{zz} ("tensor thirds": m = +1-, 0-, −1-rich), i.e. m = 0 bunches from the source	Unpolarized (helicity-averaged)	$\delta A_{zz} \approx 1\times10^{-4}$ per x-bin (combined over Q^2 and the three energies) at x < 0.05 (7×10^{-5} at $P_{zz} = 0.8$) against scenario $\lvert A_{zz}\rvert = (2$–$10)\times10^{-3}$ (HERMES-like b_1 with the 0.29 dilution) or ten times smaller (convolution): both ≥ 10σ per bin at x < 0.05. At x > 0.3, $\delta A_{zz} \approx 10^{-3}$ against a signal $\leq 2\times10^{-3}$ — JLab's domain	Relative luminosity between fills at the 10^{-4} level (a 10^{-4} offset biases A_{zz} by 1×10^{-4}; the luminosity-corrected estimator removes it exactly); tensor polarimetry, which needs dedicated development [1]
Δ via cos 2φ (^6Li; ^7Li eligible)	**Tensor** P_{zz} with the alignment axis **transverse** (the native vertical direction); fills of opposite P_{zz} sign flip the modulation — the decisive systematic check	Unpolarized	$\delta A = (1.5$–$4.5)\times10^{-4}$ per sweet-spot bin (x ≈ 0.02–0.14, Q^2 = 1.1–14 GeV^2) against moment-constrained amplitudes of $(0.7$–$1.2)\times10^{-2}$, ≥ 21σ; a 3×10^{-3} flat amplitude stays 3–10σ per bin even at $P_{zz} = 0.3$; at the reconstructed level (25% hadronic-y resolution, two-state ratio) $\delta\hat{A} = (1.0$–$3.0)\times10^{-4}$ with bin purities 0.64–0.68 — the 25% being an absolute 50–125 MeV on the hadronic $E - p_z$ sum of 0.2–0.5 GeV at the sweet spots, a regime no published ePIC study covers	Spin-state-sorted ratio of m = ±1-rich and m = 0-rich bunches, alternating bunch by bunch (it cancels the detector's cos 2φ acceptance harmonic, which biases a single-state fit by ε_2/P_{zz}); hadronic-method y at the y ≈ 0.01 sweet spots; $\delta P_{zz}/P_{zz} \approx 3\%$ scale; tensor radiative corrections not yet calculated
Spectator-tagged DIS	^7Li α-tag: any polarization for tagged F_2; vector P_z with polarized electrons for tagged g_1 of the bound proton. ^6Li α-tag: tensor P_{zz} for the order-unity tagged tensor asymmetries at $p_s \approx$ 300 MeV/c [41]	As for the inclusive observable	^7Li α → Roman Pots (R = 0.857), 96–99% acceptance, optics-robust; ^6Li α beam-blind at IP6 (3–9% with the constant 0.20 GeV proton cut; ≈ 1–3% once the 10σ envelope is scaled to the α momentum) → IR-8 secondary focus	Spectator p_T resolution ~10% at 100 MeV/c [22]; charge identification of α against A/Z = 2 fragments in the Roman Pots (open); pole extrapolation in p_T [20]
Coherent intact-^6Li cos 2φ	**Tensor** $P_{zz} \geq 0.6$, **transverse**	Unpolarized	$x \lesssim 0.01$, $\lvert t\rvert < 0.2$ GeV^2; 1.1×10^7 tagged events (high-acceptance optics; $f_0 = 0.04$ scenario), best-bin $\delta A = 1.9\times10^{-3}$ (5σ floor at a 1.0% modulation); deformation-anchored $\langle a_2\rangle = 0.036$ (band 0.018–0.059; the anchor's convention makes the cos 2β coefficient $2a_2$); a flat gluonic 3×10^{-3}–10^{-2} gives 1.6–5σ, or 5–16σ with 100 fb^{-1}/u. Reconstructed level (angular envelope, 0.22 GeV at 50 GeV/u, and the slot-like pot cutout): 2.7×10^6 tagged events, $a_t = 0.117 \pm 0.007$ and $a_e = 0.010 \pm 0.001$ in year 1 (background-free sample by assumption)	High-acceptance optics (13.5% for the constant 0.20 GeV cut; 67% / 9% / 10^{-8} at 20.5 / 50 / 137.5 GeV/u for the angular envelope, so the low-energy configuration) or IR-8; Z-identification against α+d (both R = 1); B0 calorimeter veto of the 3.56 MeV de-excitation γ

What lithium alone provides is clearest species by species. Of the ions the EIC can polarize, only the deuteron and the lithium isotopes have $J \geq 1$, so b_1 and Δ do not exist for protons or ^{3}He at all; among $J \geq 1$ nuclei ^{6}Li is the only $A > 2$ case, and the lattice indication that the deuteron's exotic glue is suppressed [11] is what makes a bound nucleus, rather than the deuteron, the likelier place for a discovery [1]. ^{7}Li is the only bound polarized proton within reach of any collider; ^{6}Li and the deuteron supply the isoscalar companions, so the three together separate the u- and d-quark spin modification — an inference of this program, not yet in the literature. For tagging, ^{7}Li gives the only light-ion spectator that is a spin-0, tightly bound nucleus landing in the Roman-Pot window today, leaving a struck $A = 3$ cluster. In the coherent channel the deuteron's deformation term dominates ($Q = +0.286$ fm^2), ^{7}Li is the deformed reference ($Q = -4.00$ fm^2), and ^{6}Li ($Q = -0.081$ fm^2) is the deformation null in which a flat-in-t modulation can only be exotic glue.

OBSERVABLE	P	D	^{3}He	^{6}Li	^{7}Li
g_1^A, polarized EMC	free reference	isoscalar pair, weakly bound; D-wave depolarization of a few per cent [20]	neutron (P_n = 0.86–0.88) at A = 3	isoscalar pair, P = 0.85, in the α field	**the bound polarized proton**, P_p = 0.87, at normal density
b_1	—	HERMES; EIC deuterons a later addition	—	**the only A > 2 case**; ^{6}Li − d isolates the α field	rank-2 analog, no published formalism
Δ (exotic glue)	— (J = ½)	allowed; lattice suggests suppression [11]	—	**allowed, bound**; spin-1 like the φ meson of the lattice signal [10]	allowed; J = 3/2 as in the bag estimate [9]
Spectator tagging	—	p, n spectators (off-momentum detectors, ZDC)	p spectators into the Roman Pots; double tag [21]	α beam-blind → IR-8; tagged tensor asymmetries O(1)	**α into the Roman Pots now**; struck A = 3 cluster
Coherent tensor cos 2φ	—	deformation term dominant [23]	—	**deformation null** (|Q| 3.5× smaller than d, opposite sign)	deformed reference

What an unpolarized lithium beam already gives. Three things, none of them the driver. First, nuclear parton distributions at A = 6 and 7: the high-x EMC ratios of ^{6}Li and ^{7}Li are in the target list of JLab E12-10-008 [54], motivated by the E03-103 finding that the ^{9}Be EMC effect follows the local density of its α clusters rather than its average density [53] — for ^{6}Li and ^{7}Li the local-density picture predicts an EMC effect well below the other models [54] — and the EIC adds the low-x shadowing region. Second, cluster virtuality: the α-tagged F_2 of the triton (^{7}Li) or deuteron (^{6}Li) cluster versus spectator momentum measures the α–t and α–d momentum distributions that VMC predicts ($N_{\alpha t} = 1.00$ in a relative p-wave, $N_{\alpha d} = 0.86$) [17], the collider counterpart of the JLab tagged-EMC program [48]. Third, coherent diffraction on an α-clustered nucleus, whose t-distribution images the gluon density of the cluster, as studied for ^{7}Li at IR-8 [27]. Polarization turns the same cluster structure from an object of study into a tool: b_1, Δ, g_1^A and the tagged tensor asymmetries all vanish or are undefined without it (§1.3).

2 · Deep inelastic scattering

An electron with four-momentum k scatters from a nucleus of momentum P by exchanging a virtual photon $q = k - k'$. The kinematic invariants, in the per-nucleon convention used throughout the program (energies and structure functions per nucleon, $x \leq 1$), are

$$Q^2 = -q^2, \qquad x = \frac{Q^2}{2P\cdot q}, \qquad y = \frac{P\cdot q}{P\cdot k}, \qquad Q^2 = xys, \qquad W^2 = M^2 + Q^2\frac{1-x}{x}$$

where x is the momentum fraction of the struck parton, y the energy-transfer fraction, and W the invariant mass of the hadronic final state. The unpolarized cross section is carried by two structure functions,

$$\frac{d^2\sigma}{dx\,dQ^2} = \frac{4\pi\alpha^2}{xQ^4}\left[\left(1 - y + \frac{y^2}{2}\right)F_2(x,Q^2) - \frac{y^2}{2}F_L(x,Q^2)\right], \qquad F_1 = \frac{F_2}{2x(1+R)}$$

with $R = \sigma_L/\sigma_T$. In the parton model, $F_2 = \Sigma_q e^2_q x[q(x)+\bar{q}(x)]$: the cross section measures quark distributions weighted by squared charge. For a nucleus we use F_2^A/A, so rates are quoted against luminosity per nucleon. Polarization introduces additional structure functions beyond F_1 and F_2; their number and character depend on the target spin.

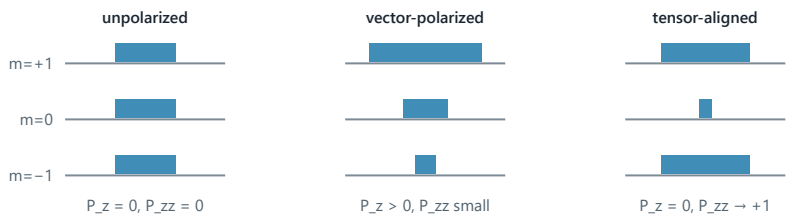
3 · Spin-1 targets

A spin-½ target contributes two spin structure functions, g_1 and g_2. The principal observable is g_1, the helicity-weighted quark distribution, measured through the double-spin asymmetry $A_{/\!/} \approx D \cdot g_1/F_1$ (D is a kinematic depolarization factor) with both beams longitudinally polarized and the electron helicity reversed bunch by bunch.

A spin-1 target carries additional structure. Its spin state along a chosen axis is specified by three populations p_+, p_0, p_- of the projections m = +1, 0, −1, and two moments characterize the fill:

$$P_z = p_+ - p_- \in [-1, 1], \qquad P_{zz} = p_+ + p_- - 2p_0 = 1 - 3p_0 \in [-2, 1]$$

P_z is the familiar vector polarization. P_{zz}, the *tensor* or alignment polarization, is specific to spin ≥ 1: it distinguishes spin along ±z from spin in the transverse plane (m = 0), and it is nonzero even when equal numbers point up and down. An unpolarized beam has populations (⅓, ⅓, ⅓) and both moments vanish; a pure m = 0 beam has $P_z = 0$ but $P_{zz} = -2$.



Hoodbhoy, Jaffe, and Manohar showed [2] that a spin-1 target has, beyond F_1, F_2, g_1, g_2, four tensor-polarized structure functions b_1–b_4, and Jaffe and Manohar identified one more, the double-helicity-flip function Δ [3]. At leading twist the tensor sector reduces to b_1 (with $b_2 = 2x\, b_1$ as its Callan–Gross partner) and Δ. For unpolarized electrons on a target whose spin axis makes angle θ_m with the beam, the cross section for each pure m state is [2,3]

$$\frac{d\sigma^{(m)}}{dx\,dy\,d\phi} = \frac{2y\alpha^2}{Q^2}\left[F_1 + \tfrac{2}{3}a_m b_1 + \frac{1-y}{xy^2}\left(F_2 + \tfrac{2}{3}a_m b_2\right) - \frac{1-y}{y^2}\, c_m \sin^2\theta_m\, \Delta(x, Q^2)\cos 2\phi\right]$$

$$a_m = \tfrac{1}{4}\, c_m(3\cos^2\theta_m - 1), \qquad c_m = 3|\lambda_m| - 2 \;\rightarrow\; (1,\ -2,\ 1) \text{ for } m = (+1,\ 0,\ -1)$$

Sign convention. The b_1, b_2 terms are written in the transcription used throughout the program. In the Hoodbhoy–Jaffe–Manohar convention as used by HERMES and by Cosyn et al. [26] (their Eq. 27: $A_{zz} = -(2/3)\, b_1/F_1$ for the axis along the photon, with the A_{zz} estimator of §4.1) the tensor-rate terms carry the opposite sign, so that the m = 0 state has the larger rate for $b_1 > 0$. Only the sign of A_{zz} relative to b_1 is affected — the magnitudes, the errors and the Δ term are not (docs/code_review_2026-08-25.md, item G1).

The weights (1, −2, 1) organize all tensor observables: the m = ±1 states enter together and against m = 0, precisely the combination measured by P_{zz}. Averaging the formula over a fill with populations p_m replaces every c_m by a factor P_{zz}. Two experimental handles follow. With the spin axis *longitudinal*, the rate difference between fills gives $A_{zz} \propto b_1$; with the spin axis *transverse* ($\theta_m = 90°$), the φ distribution of the scattered electron acquires a cos 2φ modulation proportional to Δ. Both require tensor polarization; neither requires polarized electrons.

^{7}Li, with spin 3/2, has four populations and three moments (vector, tensor/quadrupole, and octupole). Its rank-0 and rank-1 sectors are exact analogs of the spin-½ case, and host the polarized EMC measurement; its rank-2

sector is the formal analog of the spin-1 tensor sector, for which no structure-function basis has been published — an open theory item the program tracks.

4 · The measurement program

4.1 · The tensor structure function b_1

In parton language, b_1 compares quark distributions in an aligned versus a transverse nucleus [2]:

$$b_1(x) = \frac{1}{2}\sum_q e_q^2\left[q^{(0)}(x) - \frac{q^{(+1)}(x)+q^{(-1)}(x)}{2}\right], \qquad \int_0^1 b_1(x)\,dx = 0 \ \text{(Close – Kumano, no tensor sea)}$$

where $q^{(m)}$ is the quark distribution in a target with spin projection m. For a collection of free spin-½ nucleons, or for a pure S-wave bound state, all m states are equivalent and $b_1 = 0$. A nonzero b_1 therefore measures directly how binding modifies the partonic content: D-wave admixture, pion exchange, hidden-color components, or a tensor-polarized sea (which would violate the Close–Kumano sum rule [4]).

The only existing measurement is HERMES on the deuteron [5]: A_{zz} up to 2×10^{-2} over $0.01 < x < 0.45$ and $b_1(x = 0.012) = 0.112 \pm 0.055 \pm 0.028$, unexpectedly large at low x, with a zero crossing near $x \approx 0.2$ and a sum-rule integral 2σ from zero. Standard convolution calculations predict $|b_1| < 10^{-3}$ at $x \gtrsim 0.2$, an order of magnitude below the data [6], whereas a model with pions and a hidden-color probability of only 0.15% reproduces it [7]. The tension is unresolved; JLab E12-13-011 will remeasure the region $0.16 < x < 0.49$ with a solid ND_3 target [8], and the EIC adds the low-x region, where HERMES observed the large signal, together with the Q^2 lever arm.

Measurement procedure. The beam is delivered in fills of different tensor polarization ("tensor thirds"). With the estimator $(N_+ + N_- - 2N_0)$ over matched luminosities, all m-independent physics cancels, and the statistical uncertainty on A_{zz} is $\sqrt{(2/N)}/P_{zz}$. ^{6}Li would provide the first b_1 of any $A > 2$ nucleus, for which no prediction exists, and the difference between ^{6}Li and the free deuteron isolates the effect of embedding in the α field. One systematic is tracked from the outset: at collider kinematics a clean b_1 extraction favors orienting the spin axis along the momentum transfer **q** rather than along the beam [26].

4.2 · $\Delta(x,Q^2)$: exotic glue

The cos 2φ term in the master formula corresponds to the virtual photon flipping its helicity by two units between absorption and emission in the forward Compton amplitude. A spin-½ particle cannot absorb two units of helicity, so neither quarks nor bound nucleons generate Δ at leading twist; gluons can. Δ is the structure function that the gluon transversity distribution, an interference of gluon helicity states existing only in targets with $J \geq 1$, generates at order α_s through a photon–gluon convolution [3,9]. A nonzero Δ in a nucleus is therefore direct evidence for glue not inherited from individual nucleons, hence the name "exotic glue". It has not been measured for any target; the one prior experimental document is a JLab letter of intent [13].

The measurement uses unpolarized electrons on a tensor-polarized nucleus whose alignment axis lies in the transverse plane, and fits the azimuthal distribution about that axis. Averaged over the fill, the modulation amplitude and its inversion back to Δ are

$$A^{\cos 2\phi} = -\frac{1-y}{y^2}\frac{\Delta}{D_\phi}, \qquad D_\phi = F_1 + \frac{1-y}{xy^2}F_2$$

The expected size spans per-mille to percent. The only absolute normalizations available are moment estimates: a bag-model calculation by Sather and Schmidt for the spin-3/2 Δ^{++} baryon, $\int x\,\Delta\,dx = -0.012\,\alpha_s$ [9], and lattice-QCD calculations by Detmold and Shanahan of gluon transversity in the φ meson (clearly nonzero, albeit at an unphysical pion mass) [10], with NPLQCD finding the analogous matrix element roughly ten times suppressed in the deuteron [11]. Percent-level asymmetries appear only in proton–deuteron Drell–Yan under the maximal assumption $\Delta_T g = \Delta g$, described by its authors as optimistic [12]. No calculation exists for ^{6}Li or any $A > 2$

nucleus, and the literature brackets the nuclear Δ in both directions: suppressed, as in the lattice deuteron, or enhanced by binding in analogy with the EMC effect [1]. The projection report transfers the bag moment to ^{6}Li as a labeled scenario, shows that one EIC year measures the amplitude at ≥ 20σ per bin, and details the caveats.

The coherent channel. The same modulation can be measured in coherent diffraction, e ^{6}Li → e′ X ^{6}Li(g.s.), in which the nucleus remains in its ground state and recoils intact. There the cos 2φ modulation mixes two mechanisms: the nuclear *shape*, in which a deformed gluon density modulates the diffractive slope (the mechanism computed for polarized deuterons in the only existing calculation of this kind [23]), and exotic glue proper, which persists at small x where it connects to the elliptic-gluon program [24]. ^{6}Li discriminates between the two because its quadrupole moment is anomalously small (Q = −0.0806 fm², roughly 50 times smaller than that of ^{7}Li): the shape term is bounded and linear in t, so a flat-in-t modulation cannot be a shape effect. Detection of the intact recoil is itself nontrivial: the nucleus keeps the beam rigidity, as any coherent recoil does, and emerges within ~2 mrad of the beam, visible only in the Roman-Pot transverse-momentum tail at IP6 or at the secondary focus of a second interaction region [25,27]; the companion report treats this in detail.

4.3 · The polarized EMC effect: g_1 of a bound proton

The EMC effect — the ~10–20% suppression of F_2 per nucleon in a nucleus at $0.3 \lesssim x \lesssim 0.7$ — has been measured for forty years without a settled explanation. Its spin analog has never been measured at all: does the *spin* structure of a bound nucleon change more, less, or the same as its momentum structure? The observable is the polarized EMC ratio

$$\Delta R_A(x) = \frac{g_1^A(x)}{P_p\, g_1^p(x) + P_n\, g_1^n(x)}$$

where P_p, P_n are the effective nucleon polarizations from nuclear structure. ^{7}Li is well suited: the variational Monte Carlo spin-projected nucleon numbers of Wiringa et al. give P_p = +0.87 and P_n = −0.04 (0.866 and −0.037 as quoted in the JLab proposal) [17,45], so to a good approximation polarized ^{7}Li is one polarized proton plus six spectators, with computable corrections. Two classes of models make sharply different predictions: Cloët–Bentz–Thomas (mean-field, NJL) obtain a polarized EMC effect about twice the unpolarized one [14], whereas Tronchin–Matevosyan–Thomas obtain equal effects [15]. A measurement at the 5% level discriminates between them, and the two map onto the competing mean-field and short-range-correlation pictures of the EMC effect itself. A handle unique to the EIC is the Q^2 evolution of g_1, driven by the polarized gluon distribution; the predicted *gluon-spin* EMC effect is larger still, with ^{7}Li identified as the most promising case [16]. JLab E12-14-001 will measure ^{7}Li at x > 0.06 [45]; the EIC extends the coverage to x < 0.06 and adds the Q^2 arm, in collider kinematics free of target-fragment absorption.

For this observable both beams are polarized longitudinally and the electron helicity flips bunch-by-bunch; the error scales as $1/(P_e P_z \sqrt{N})$, so it uses the vector polarization the source delivers most readily.

4.4 · Spectator-tagged DIS

The cluster description of ^{6}Li is quantitative (§1.2): the α is inert, and the deuteron-like pair carries the nuclear spin with a polarization of 0.85–0.9 in six- and three-body calculations [17,18,19], the 0.87 adopted by E155 [43] being the value the program uses. When the α spectator is detected (it carries beam velocity into the far-forward spectrometers), the struck object is identified as the deuteron cluster, and the ^{6}Li beam acts as a source of polarized deuterons at controlled virtuality. The formal framework is spectator tagging (Cosyn–Weiss [20]): the spectator momentum is measured and extrapolated in its transverse component to the free-nucleon (here free-cluster) pole, recovering free-particle structure with binding effects removed by construction rather than modeled. Tagged *tensor* asymmetries are predicted to reach O(1) at spectator momenta of ~300 MeV/c, where the D-wave dominates [41] — far larger than the inclusive asymmetries above. The benchmark for the method is the e+^{3}He double-tagging study of Friščić et al. [21], in which tagging improved on effective-polarization unfolding by a factor of 2–3, and by a factor of 10 at low x.

Far-forward routing determines where each isotope's tag fragment is detected, and it follows from magnetic rigidity alone: a beam-velocity fragment with mass-to-charge ratio A/Z bends in the hadron optics according to $R = (A/Z)_{frag}/(A/Z)_{beam}$.

BEAM	TAG FRAGMENT	R	DESTINATION AT IP6
^{7}Li (A/Z = 7/3)	α	0.857	Roman Pots main window (R ∈ 0.60–0.95) — works now
^{7}Li	t	1.286	over-rigid, no coverage
^{6}Li (A/Z = 2)	α or d	1.000	beam rigidity: inside the beam envelope except the p_T tail
^{6}Li	p / n	0.50 / –	Off-Momentum Detectors / ZDC (veto handles)

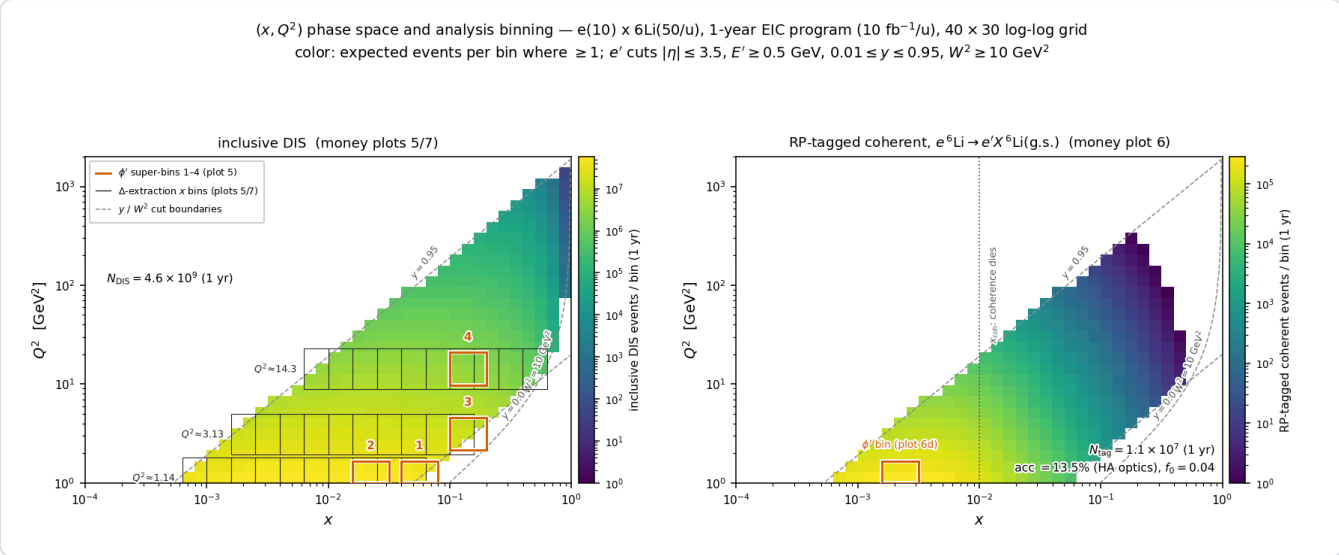
The tagging program therefore begins with ^{7}Li at IP6, while ^{6}Li α-tagging (and the coherent intact-^{6}Li channel, which faces the same rigidity problem) motivates the secondary focus of the second interaction region [27]. This asymmetry between the isotopes was itself a finding of the simulation program.

5 · Rates and statistical precision

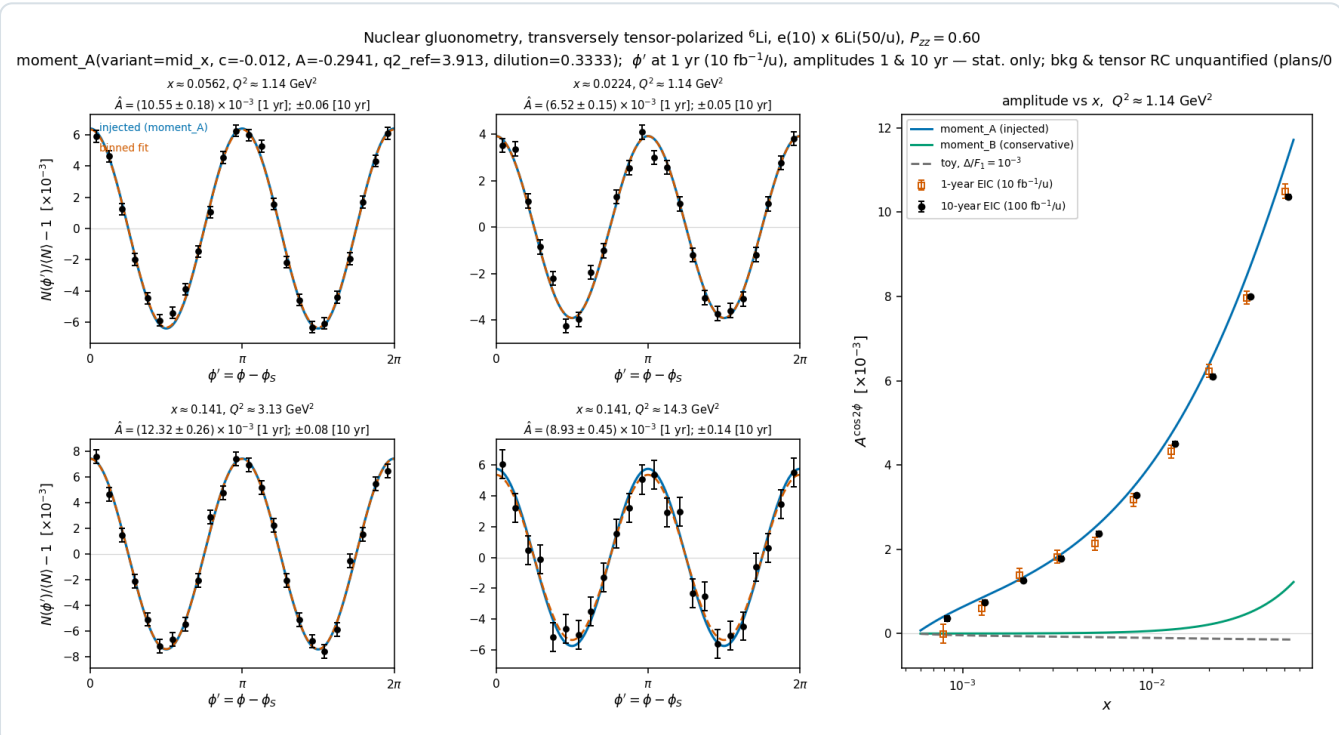
All projections use luminosity per nucleon, with 10 fb^{-1}/u — 30 weeks of operation at 10^{33} cm^{-2} s^{-1} with 60% efficiency in the Yellow Report's accounting [25], rounded here to "one EIC year" — and 100 fb^{-1}/u for the ten-year program, and in-ring polarization placeholders $P_z = 0.7$ and $P_{zz} = 0.6$ (ring transport from the source values is an open accelerator question). The statistical uncertainties behind every money plot are

$$\delta A_{\parallel} = \frac{1}{P_e P_z \sqrt{N}}, \qquad \delta A_{zz} = \frac{\sqrt{2/N}}{P_{zz}}, \qquad \delta A^{\cos 2\phi} = \frac{\sqrt{2/N}}{P_{zz}} \text{ (single state)}, \qquad \delta \hat{A} = \frac{2}{P_+ - P_0}\sqrt{\frac{2}{N}} \text{ (two states)}$$

The factor √2 in the tensor uncertainties follows from $\langle \cos^2 2\varphi \rangle = ½$ (equivalently, from the three-fill estimator variance); the polarization appears in the denominator because the fill dilutes the physical asymmetry. These formulas, and their behavior under realistic φ-acceptance holes (where a binned fit replaces the naive moment), are validated end to end by pseudo-experiments with the polligen event generator. Tensor observables enjoy a structural advantage in systematics when the m = ±1-rich and m = 0-rich states are measured with the same detector and compared bin by bin: in their spin-state ratio the φ acceptance cancels and relative-luminosity errors enter only as a constant (a single-state fit, by contrast, measures the detector's own cos 2φ acceptance harmonic divided by P_{zz}), and the two-state pattern $P_+ = +0.6$, $P_0 = -1.2$ is 1.5× more precise than a single state at the same source purity — provided the states alternate bunch by bunch, since a 10^{-3} difference of the acceptance harmonic between them fakes half the signal (reconstruction-chain report §3). What remains is the polarimetry scale ($\delta P_{zz}/P_{zz} \approx 3\%$) and radiative corrections, which for tensor observables have not yet been calculated.



Phase-space coverage. Expected events per (x, Q^2) bin in one EIC year at e(10 GeV) × ^{6}Li(50 GeV/u). Left: inclusive DIS (4.6×10^9 accepted events) with the bins of the cos 2φ analysis overlaid; the measurement is limited by polarization, acceptance, and systematics rather than by statistics. Right: the coherent intact-^{6}Li channel (1.1×10^7 tagged events with the constant 0.20 GeV cut; 2.7×10^6 with the angular envelope and slot cutout of the reconstruction-chain report), confined to x $\lesssim$ 0.03 because coherence requires the photon to resolve the whole nucleus, a condition that fails once x $\gtrsim$ 0.01.



Projected extraction. cos 2φ′ pseudo-data (one EIC year, P_{zz} = 0.6) in the four highest-significance (x, Q^2) bins, with the injected moment-constrained Δ model and the binned fit; right, the extracted amplitude versus x against alternative model readings. The projected precision distinguishes between the models rather than only from zero. Assumptions and caveats are detailed in the companion report; its reconstructed-level version (money plot 5R, reconstruction-chain report §7) uses the spin-state ratio in reconstructed bins.

6 · Further reading

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Appendix A · Modification history

The main text states the current content only; this appendix records what changed and when (the repository history carries the full diffs).

DATE	REVISION
2026-08-17	First version: the DIS and spin-1 formalism developed from first principles, the four flagship measurements (b_1, Δ, the polarized EMC effect, spectator-tagged DIS) with their procedures and statistical reach; academic-register line edit the same day.
2026-08-21	§1 rewritten — why lithium as a source and as a target, why polarization is indispensable, fixed target versus collider — fact-checked against primary sources; §1.5 added (what each observable requires of the beam, and what lithium alone provides).
2026-08-25	References verified against the local copies in refs/: [28] corrected to E. Hamwi and G. H. Hoffstaetter, *Polarization transmission in the Electron-Ion Collider's Hadron Storage Ring* (J. Devlin is acknowledged, not an author; the title previously cited does not exist); [41] corrected to *Semi-inclusive DIS on a polarized spin-1 target*, Parts I/II, with the verbatim order-unity statement.
2026-08-25	§1.5 table: the spin-state ratio and bunch-by-bunch alternation as the Δ requirement, hadronic-method y at the sweet spots, the reconstructed-level 5R errors; the ^{6}Li α-tag 3–9% identified as the constant-cut value (≈ 1–3% with the angular envelope); the coherent row with the angular envelope and the reconstructed-level 6R numbers; sign convention of the b_1, b_2 terms in §3; the two-

	state error formula and the systematics paragraph of §5 (a single-state fit measures the acceptance harmonic); Yellow Report accounting with its 60% efficiency; δA_{zz} binning stated; [24] cited; figure captions point to the reconstructed-level versions.
2026-08-25	Modification history moved to this appendix; the main text restated as current content.

Educational companion to `reports/cos2phi_money_plots_report.html` (projections and caveats), `reports/reconstruction_chain_report.html` (the measurement chain and the reconstructed-level results) and `plans/01_findings_physics_case.md` (sourced physics case). Generated by `reports/build_report.py`; display math typeset at build time (matplotlib mathtext, STIX), figures from the evgen scripts. Projections use scenario structure functions with explicit bands.